

From Epsilon-Greedy to Boltzmann Exploration in Deep Q-Learning

Report on the *Epsilon greedy to Boltzmann* Project

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1 Project overview

The repository `sapankowski/Epsilon-greedy-to-Boltzmann` is an experimental reinforcement-learning project focused on exploration strategies in Deep Q-Networks (DQN). The main idea is to compare standard epsilon-greedy exploration with Boltzmann-style exploration and with hybrid schedules that gradually move from one regime to the other.

The central research question is whether exploration should be mostly random at the beginning of training and increasingly value-dependent later. This is a natural hypothesis in DQN. Early in training, Q-values are noisy and poorly calibrated, so sampling actions directly from a softmax over Q-values may be misleading. Epsilon-greedy exploration is robust in this phase because it does not depend strongly on the numerical scale of the current Q-function. Later in training, however, the learned Q-values can contain useful information about action quality. Boltzmann exploration can then be preferable because it samples actions proportionally to their estimated value rather than treating all non-greedy actions equally.

The project evaluates this idea on CartPole, LunarLander, and Breakout. CartPole is a fast sanity-check environment, LunarLander is a more demanding continuous-state control task with discrete actions, and Breakout represents the Atari/image-observation path. The figures included below summarize the reported learning curves and final evaluation returns for all three tasks.

2 DQN background

The learning algorithm is based on DQN, where a neural network approximates the action-value function $Q_\theta(s, a)$. For a transition (s, a, r, s') , the standard DQN target is

$$y = r + \gamma \max_{a'} Q_{\theta^-}(s', a'), \quad (1)$$

where θ denotes the online network and θ^- denotes the target network. The target network stabilizes learning by making the bootstrap target change more slowly than the online network.

The project also supports Double DQN. In Double DQN, the online network chooses the next action, while the target network evaluates that action:

$$y = r + \gamma Q_{\theta^-}\left(s', \arg \max_{a'} Q_\theta(s', a')\right). \quad (2)$$

This modification is intended to reduce overestimation bias, a known issue in Q-learning-based methods. In addition, the implementation uses standard DQN components such as replay memory, target-network updates, Huber loss, gradient clipping, and separate model architectures for vector observations and image observations.

3 Exploration strategies

The main experimental variable is the action-selection rule. Each strategy receives Q-values for the current state and returns a probability distribution or sampling rule over the available discrete actions. The repository compares five exploration variants:

Strategy	Mechanism	Interpretation
<code>epsilon_greedy</code>	greedy action plus uniform random exploration	stable early-training baseline
<code>boltzmann</code>	softmax over $Q(s, a)/T$	value-weighted stochastic policy
<code>annealed_epsilon_boltzmann</code>	gradual mixture of epsilon-greedy and Boltzmann policies	transition from random to value-based exploration
<code>topk_boltzmann</code>	epsilon exploration, then softmax over the top- k actions	avoids sampling clearly poor actions late in training
<code>uncertainty_switch</code>	transition depends on schedule and Q-value entropy	switches when action preferences become sharper

Table 1: Exploration variants compared in the project.

For epsilon-greedy exploration, the greedy action receives most of the probability mass, while the remaining mass is distributed uniformly:

$$\pi(a \mid s) = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{|\mathcal{A}|}, & a = \arg \max_b Q(s, b), \\ \frac{\varepsilon}{|\mathcal{A}|}, & \text{otherwise.} \end{cases} \quad (3)$$

Boltzmann exploration instead samples actions according to

$$\pi(a \mid s) = \frac{\exp(Q(s, a)/T)}{\sum_b \exp(Q(s, b)/T)}, \quad (4)$$

where T is the temperature. High temperature produces a flatter distribution; low temperature makes the distribution closer to greedy action selection.

4 CartPole results

CartPole is the simplest benchmark in the project and serves mainly as a correctness and stability test for the training pipeline. The final evaluation plot indicates that `annealed_epsilon_boltzmann` obtains the highest mean final return among the compared strategies, around 454. `uncertainty_switch` is also strong, with a mean final return around 415. Standard `epsilon_greedy` is weaker on average, and `topk_boltzmann` performs worst in this particular CartPole run.

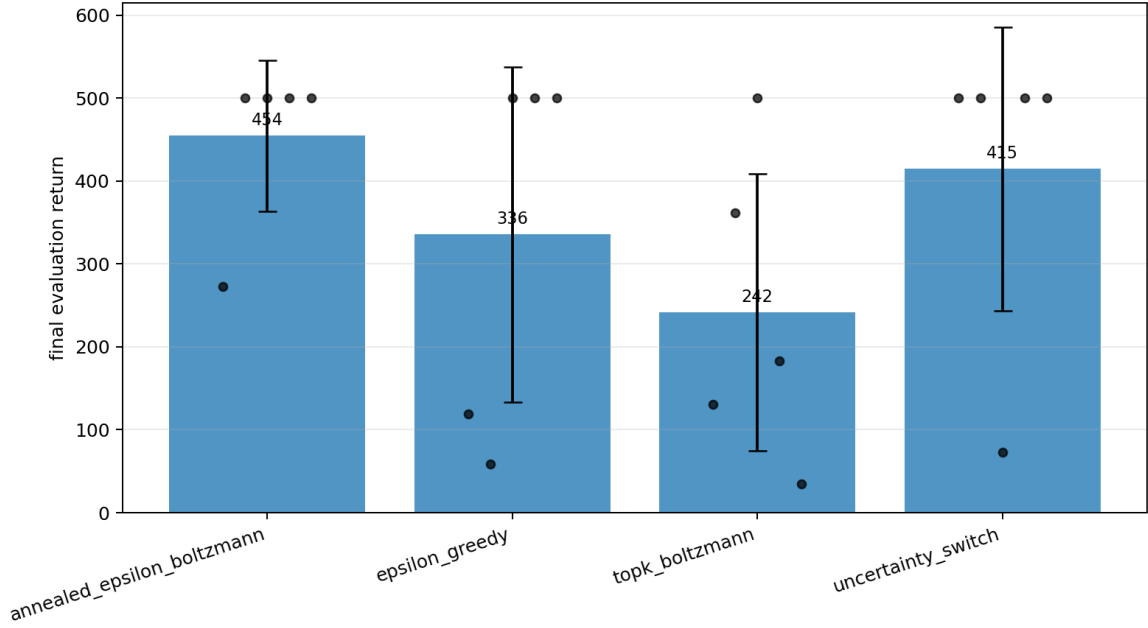


Figure 1: CartPole final evaluation return. Bars show mean final return, black points show individual runs, and error bars show variability across seeds.

The learning curve gives more context. The annealed strategy improves sharply near the later part of training and approaches the solved-threshold region. The uncertainty-switch strategy also improves strongly near the end. Epsilon-greedy learns more slowly and reaches a lower final mean. The large shaded intervals show substantial seed-to-seed variation, which is expected for small-seed reinforcement-learning experiments.

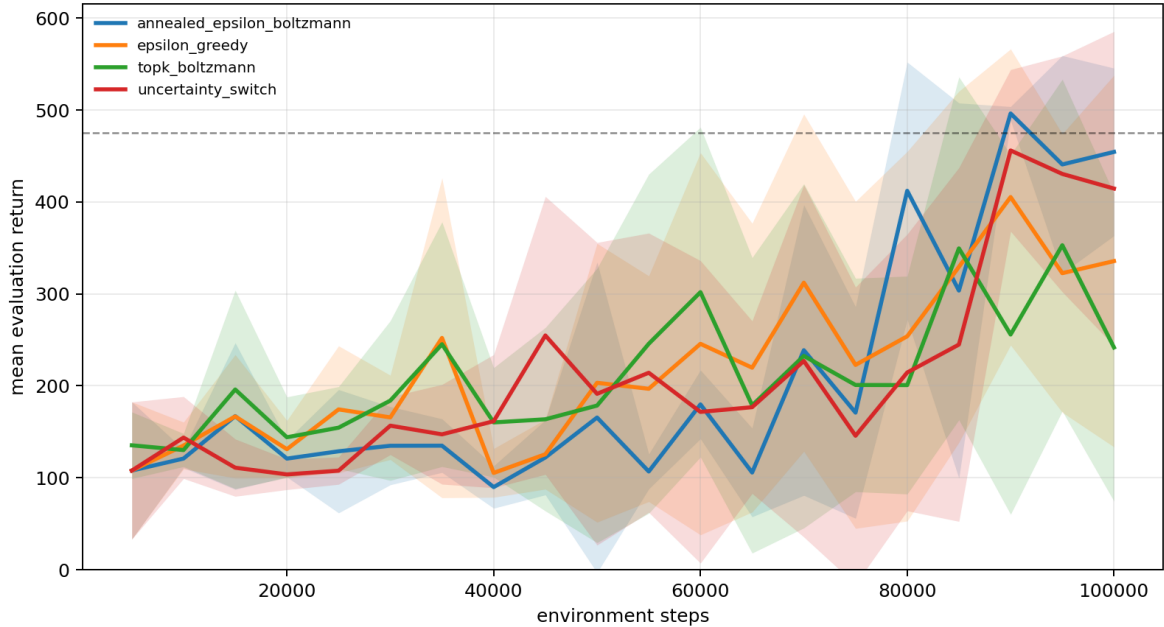


Figure 2: CartPole mean evaluation return over environment steps. The dashed horizontal line marks the approximate solved threshold.

5 LunarLander results

LunarLander is more informative than CartPole because it is harder and has a more complex reward landscape. In the final-return plot, `uncertainty_switch` and `annealed_epsilon_boltzmann` are the best-performing methods, both reaching mean final returns around 240. Standard `epsilon_greedy` and `topk_boltzmann` obtain lower means, around 193 and 188 respectively.

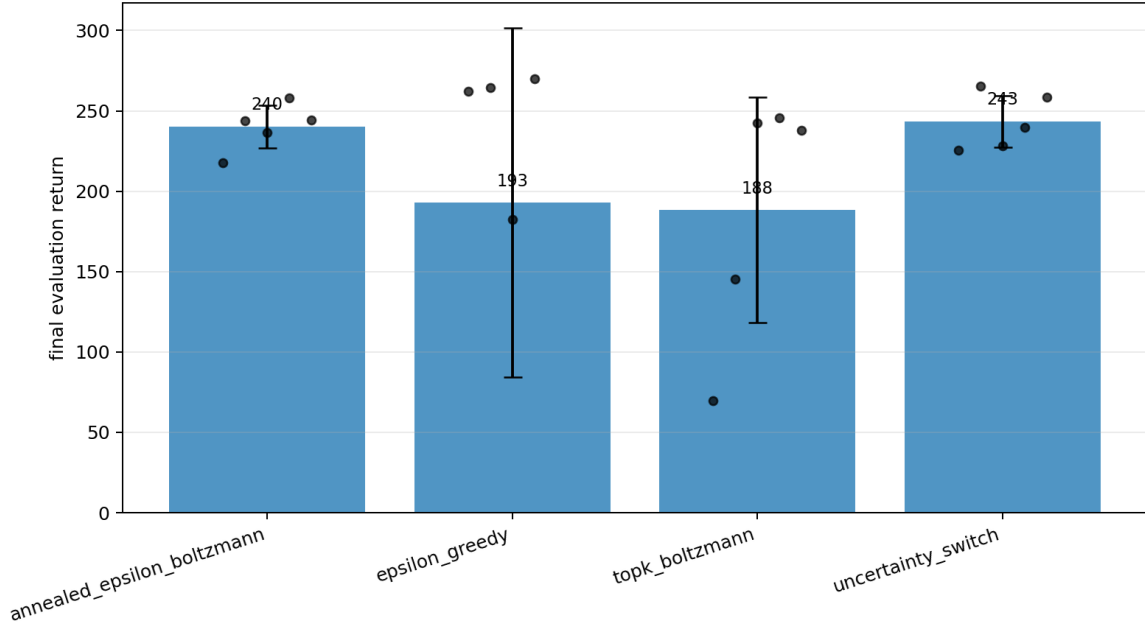


Figure 3: LunarLander final evaluation return. The best final means are achieved by the annealed and uncertainty-switch strategies.

The learning curve suggests that the annealed method is the most consistently strong strategy over much of training. It reaches and stays near or above the solved-threshold region earlier than the other methods. The uncertainty-based strategy also improves substantially, although its curve is more irregular. Epsilon-greedy improves more slowly, while top- k Boltzmann lags behind for most of the run.

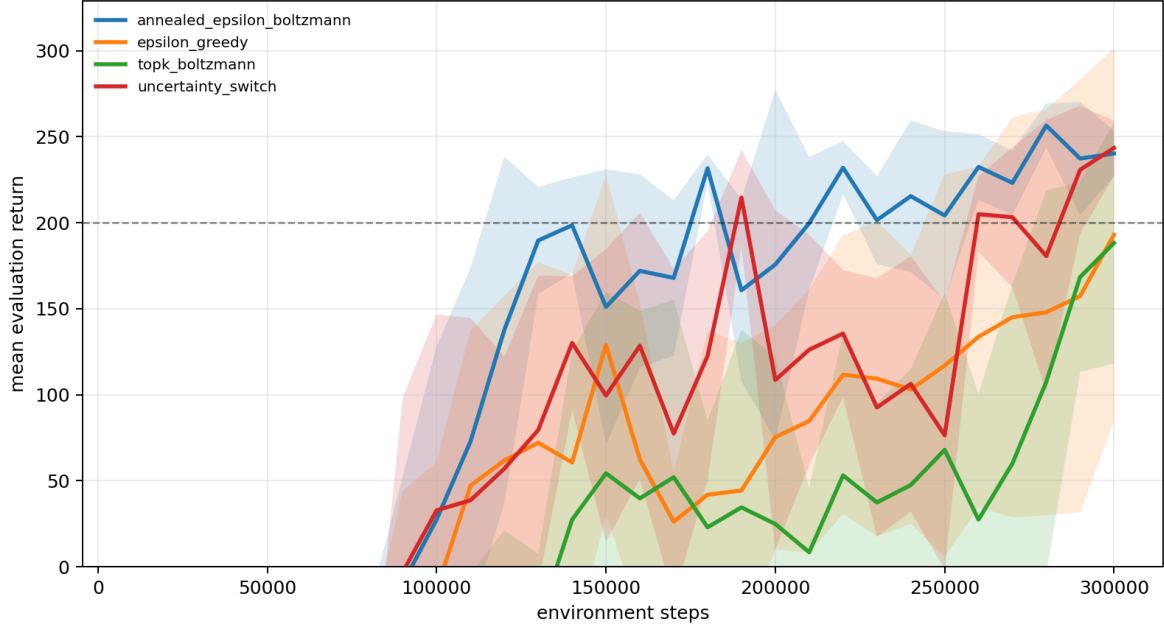


Figure 4: LunarLander mean evaluation return over environment steps. The dashed horizontal line marks the approximate solved threshold.

6 Breakout results

Breakout is the most computationally demanding environment in the project because it uses image observations and requires a convolutional Q-network. The final-return plot shows a different pattern from CartPole and LunarLander. The best final mean is achieved by the uncertainty-based strategy, around 121, followed by epsilon-greedy at around 115. Top- k Boltzmann reaches a lower final mean, while the annealed strategy performs poorly in the displayed Breakout result.

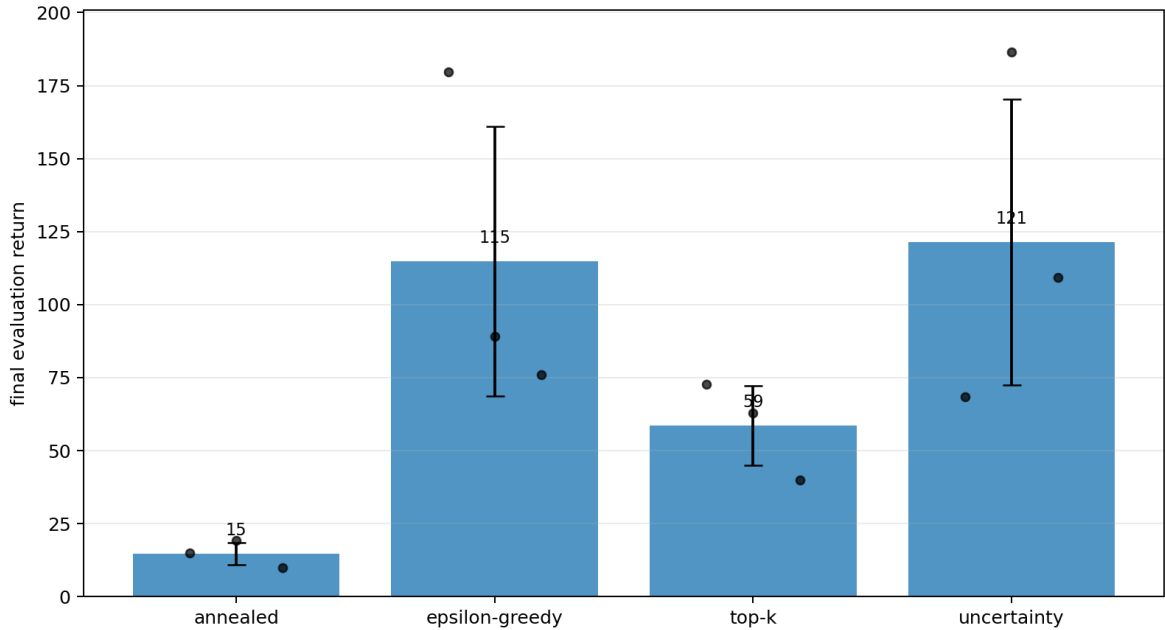


Figure 5: Breakout final evaluation return. In this run, uncertainty-based exploration and epsilon-greedy perform best, while the annealed strategy is weak.

The Breakout learning curve shows that epsilon-greedy, top- k , and uncertainty-based exploration improve over time, while the annealed curve remains much lower. This suggests that the transition schedule that works well for simpler control tasks may not transfer directly to Atari. A plausible explanation is that image-based DQN requires much longer training before Q-values become reliable enough for Boltzmann-style sampling. Therefore, moving away from robust random exploration too early may harm performance.

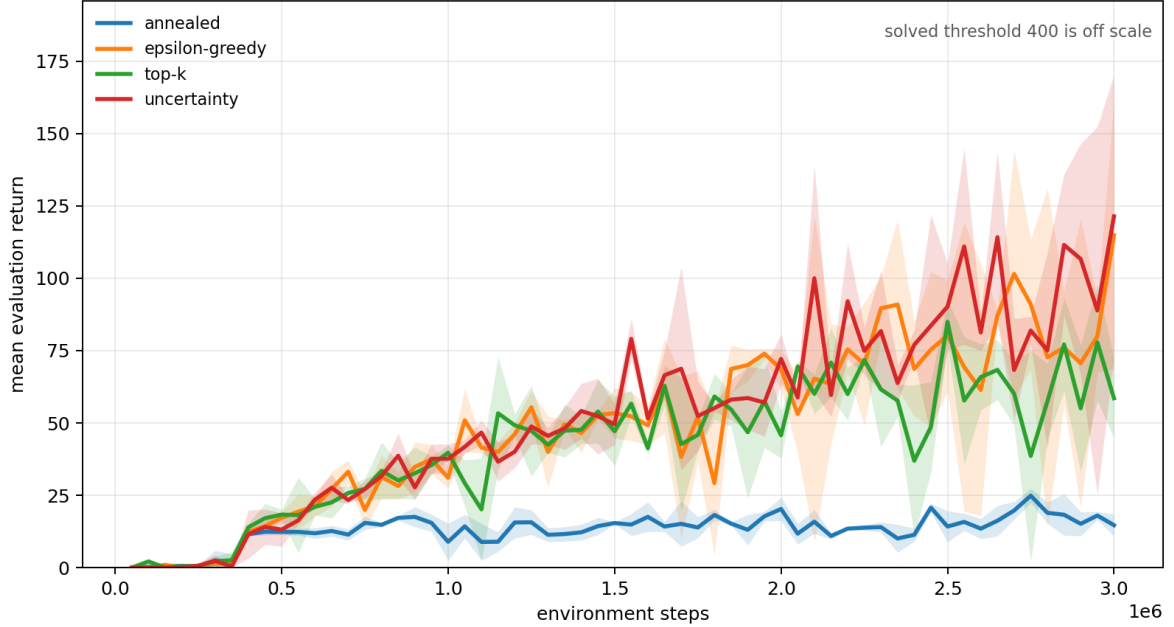


Figure 6: Breakout mean evaluation return over environment steps. The solved threshold of 400 is outside the plotted scale.

7 Discussion

The results support the main intuition of the project, but only partially. On CartPole and LunarLander, the annealed epsilon-to-Boltzmann approach is highly competitive and often better than plain epsilon-greedy. This suggests that a gradual transition from random exploration to value-aware stochastic exploration can be beneficial when the environment is relatively low-dimensional and the Q-network becomes useful within the available training budget.

However, the Breakout results are more cautious. In the displayed Atari experiment, the annealed method underperforms, while uncertainty-based exploration and epsilon-greedy are stronger. This indicates that Boltzmann-based exploration is sensitive to Q-value scale, value calibration, temperature scheduling, and the length of training. In image-based Atari environments, Q-values may remain unreliable for a long time, so a premature transition to value-dependent sampling can amplify noise rather than improve exploration.

The main limitation is statistical strength. Reinforcement-learning results can vary substantially across random seeds, and several of the plots show wide uncertainty bands. Stronger conclusions would require more seeds, longer Atari training, and ablations over temperature schedules, transition schedules, and entropy thresholds. It would also be useful to compare these methods with stronger exploration baselines such as NoisyNets, bootstrapped DQN, or count-based intrinsic rewards.

8 Conclusion

Overall, the project implements a DQN-based experimental framework and uses it to evaluate a focused exploration question: when should an agent move from random exploration to Q-value-aware stochastic action selection? The CartPole and LunarLander experiments suggest that annealing from epsilon-greedy to Boltzmann exploration can be effective in smaller control environments. The Breakout results show that the same idea does not automatically scale to image-based Atari tasks and may require a slower or uncertainty-aware transition.

The most defensible conclusion is therefore not that Boltzmann exploration universally improves DQN, but that adaptive exploration schedules are promising and environment-dependent. The repository provides a useful foundation for further experiments, especially if extended with more seeds, longer runs, and stronger uncertainty estimates.

References

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